

# 09 – Web Development

Creating the World Wide Web

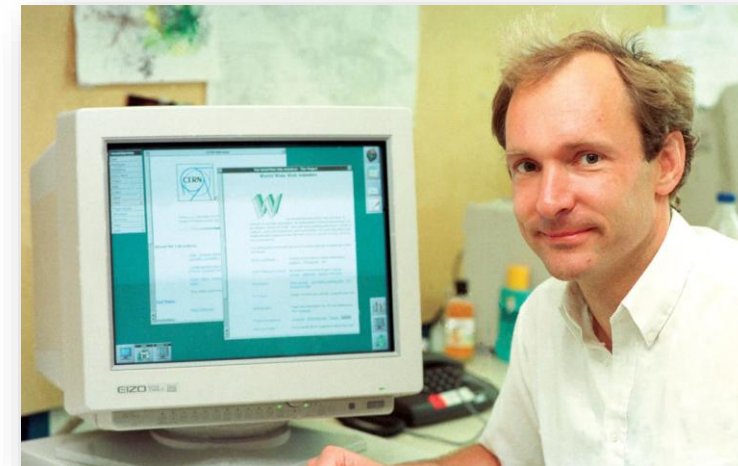
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# OUTLINE

- 1) The Web
- 2) Developing static web pages
- 3) Developing dynamic web pages
- 4) Front-end / Back-end development
- 5) Why Node.js? Why Javascript?

# 1) THE WEB

- **The World Wide Web** is a network of web servers delivering web pages through the **http protocol**
- It has been created by **Tim Berners-Lee, Robert Cailliau et al.** in 1989 at CERN, Geneva, Switzerland
- The Web is only a part of the Internet
- Web pages are written in the **HTML** language
- HTML5 is the latest version of HTML
- In a web page, a **anchor link** can redirect to another web page everywhere on the web

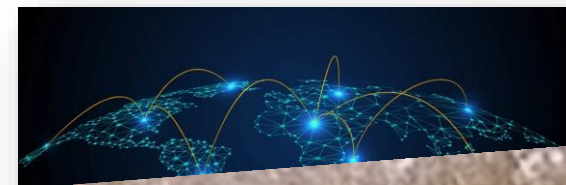


# 1) THE WEB



Images: <https://brightlineit.com/wp-content/uploads/2017/06/170613-upgrade-your-internet-connection.jpg>,  
<https://www.britannica.com/topic/World-Wide-Web>

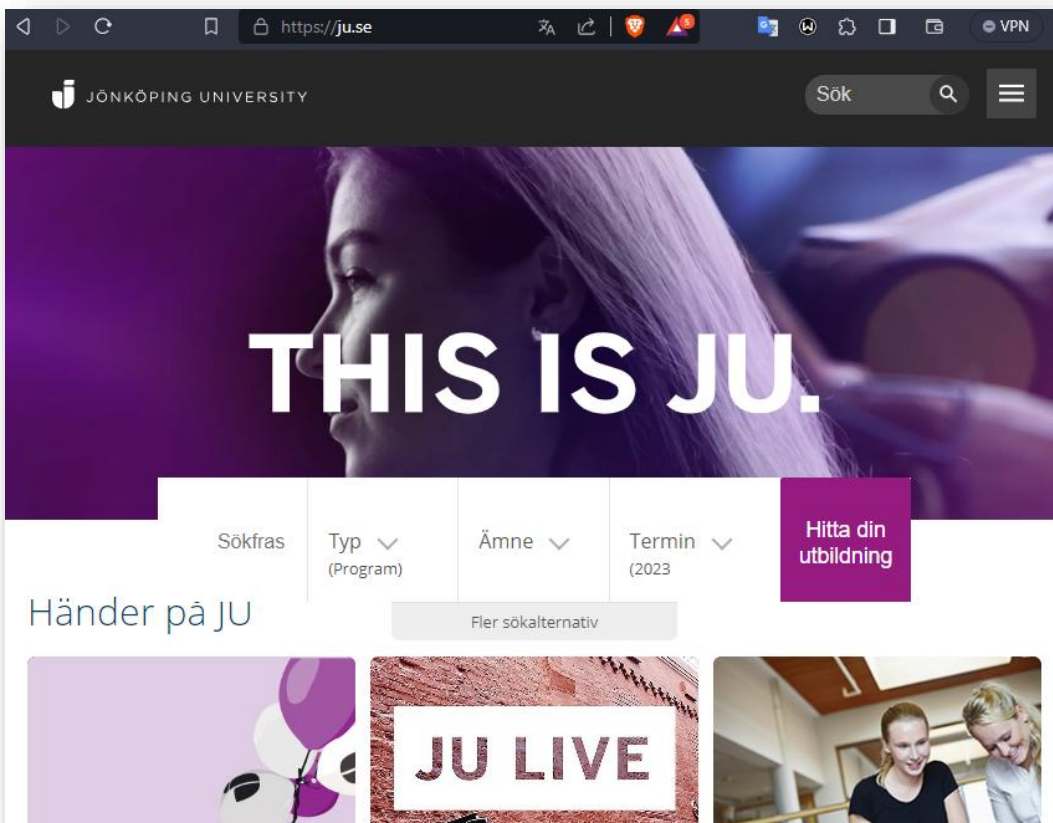
# 1) THE WEB



On a web page, a **anchor link** can redirect to another web page everywhere on the web

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<https://www.britannica.com/topic/World-Wide-Web>





```
Line wrap
1 <!DOCTYPE html>
2 <html lang="sv" class="sv-no-js sv-template-kampanjwebb">
3 <head>
4   <meta charset="UTF-8">
5   <script nonce="ab0ff6a0-4249-11ee-bd95-cd78b8da360d">(function(H){H.className=H.className.replace(/\bsv-no-
6 js\b/, 'sv-js')})(document.documentElement)</script>
7   <title>Jönköping University</title>
8   <meta name="informationOwner" content="camilla.hallgren@ju.se" />
9   <meta name="webEditor" content="Patrik.Sandberg@ju.se" />
10  <meta name="company" content="HJ" />
11  <meta name="companyName" content="Jönköping University" />
12  <meta name="companyHomepage" content="/" />
13  <meta name="publishedAt" content="2023-08-22" />
14  <meta name="featureSource" content="/fragment/feature/2023-film-for-filtrering.html" />
15  <meta name="contentgroup" content="JU" />
16  <meta name="viewport" content="width=device-width, initial-scale=1, minimum-scale=1, shrink-to-fit=no" />
17  <meta name="dcterms.identifier" content="http://ju.se">
18  <meta name="dcterms.language" content="sv">
19  <meta name="dcterms.format" content="text/html">
20  <meta name="dcterms.type" content="text">
21  <link rel="stylesheet" type="text/css" media="screen,print"
22 href="/2.b3cf64012843bc4e2880001270/1691909865088/sitevision-responsive-grids.css" />
23  <link rel="stylesheet" type="text/css" media="screen,print"
24 href="/2.b3cf64012843bc4e2880001270/1691909882931/sitevision-spacing.css" />
25  <link rel="stylesheet" type="text/css" media="screen,print"
26 href="/2.b3cf64012843bc4e2880001270/1691909882931/sitevision-spacing.css" />
```

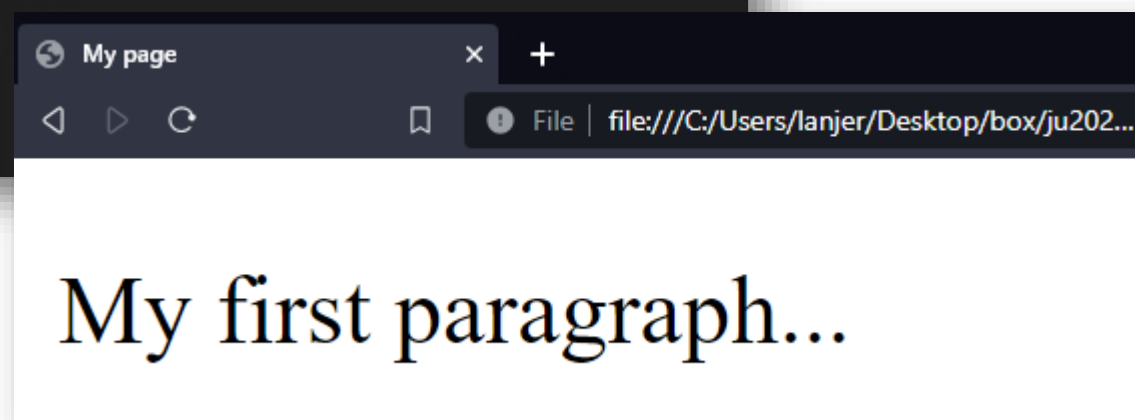
## 2) DEVELOPING STATIC WEB PAGES

- **HTML** (hyperText Markup Language) is a description language where you use **tags** to organise information
- To develop **static** web pages, you only need a **text editor** and a **browser** to see the result
- A minimal HTML5 document:

```
projects-jl > <> index.html > ...
1  <!DOCTYPE html>
2  <html lang="en">
3  <head>
4  |   <meta charset="UTF-8">
5  |   <meta name="viewport" content="width=device-width, initial-scale=1.0">
6  |   <title>My page</title>
7  </head>
8  <body>
9  |   <p>My first paragraph...</p>
10 </body>
11 </html>
12
```

## 2) DEVELOPING STATIC WEB PAGES

```
<!DOCTYPE html>
<html lang="en">
  <head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-scale=1.0">
    <title>My page</title>
  </head>
  <body>
    <p>My first paragraph...</p>
  </body>
</html>
```





- **CSS** (Cascading Style Sheet) is a language used to assign properties to HTML tags to apply styles to elements
- A minimal HTML5 document and its external CSS style sheet:

<> index.html ×

project-languages > <> index.html >  html

```
1  <!DOCTYPE html>
2  <html lang="en">
3  <head>
4      <meta charset="UTF-8">
5      <meta name="viewport" content="width=device-width, initial-scale=1.0">
6      <title>My page</title>
7      <link rel="stylesheet" href="styles-jl.css">
8  </head>
9  <body>
10     <p>My first paragraph...</p>
11 </body>
12 </html>
```

# styles-jl.css ×

project-languages > # styles-jl.css > ...

```
1  body {
2      font-family: sans-serif;
3  }
4
5  p {
6      color: ■ darkgreen;
7  }
8
```

&lt;&gt; index.html ✕

project-languages > <> index.html >  html

```
1  <!DOCTYPE html>
2  <html lang="en">
3  <head>
4    <meta charset="UTF-8">
5    <meta name="viewport" content="width=device-width, initial-scale=1.0">
6    <title>My page</title>
7    <link rel="stylesheet" href="styles-jl.css">
8  </head>
9  <body>
10   <p>My first paragraph...</p>
11 </body>
12 </html>
```

# styles-jl.css ✕

project-languages &gt; # styles-jl.css &gt; ...

```
1  body {
2    font-family: sans-serif;
3  }
4
5  p {
6    color: ■ darkgreen;
7  }
8
```

Without the style file:

My first paragraph...

With the style file:

My first paragraph...

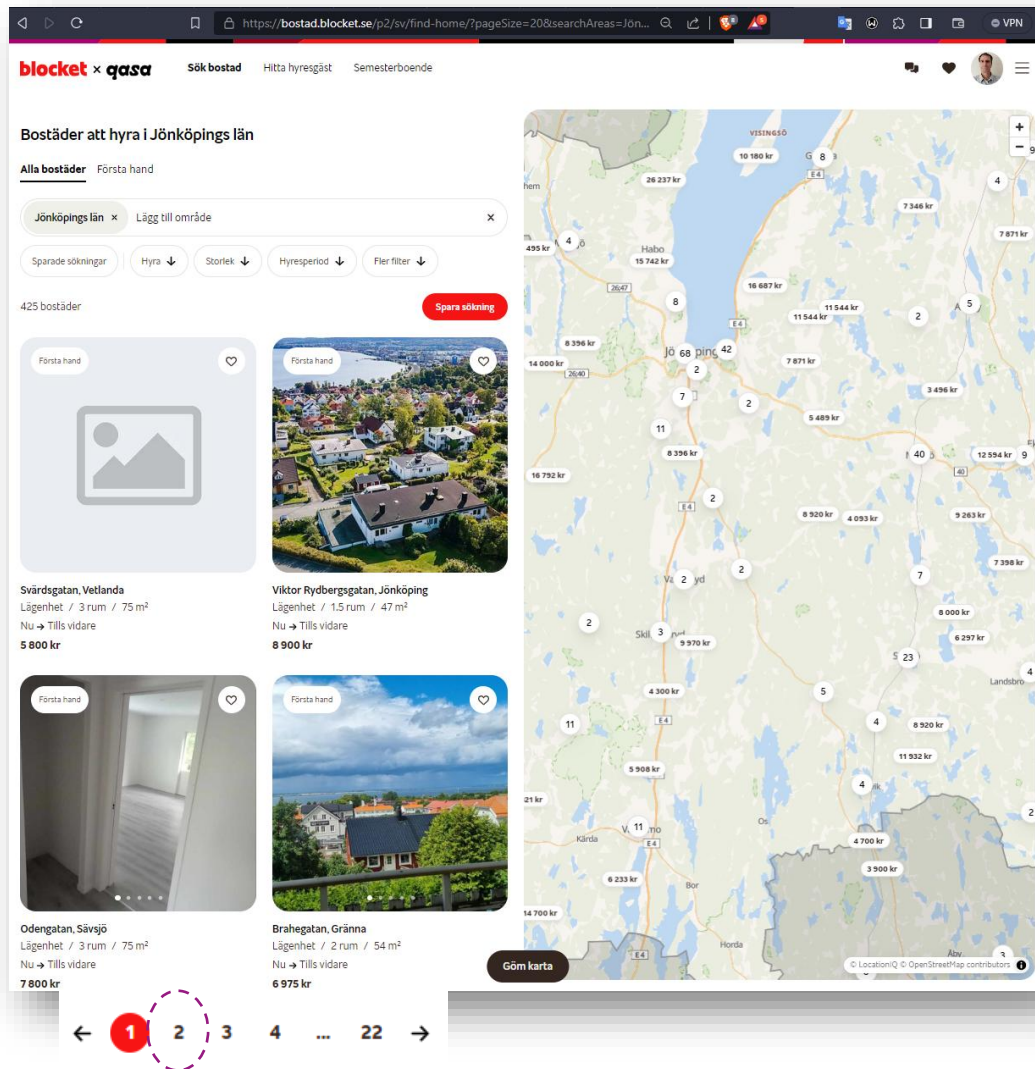
## 2) DEVELOPING STATIC WEB PAGES

- The main problem with static pages is you need to reload **all the page** each time there is a change in the information
- In order to **increase performance**, it is preferable to load only a small part of the page (the one that changes)

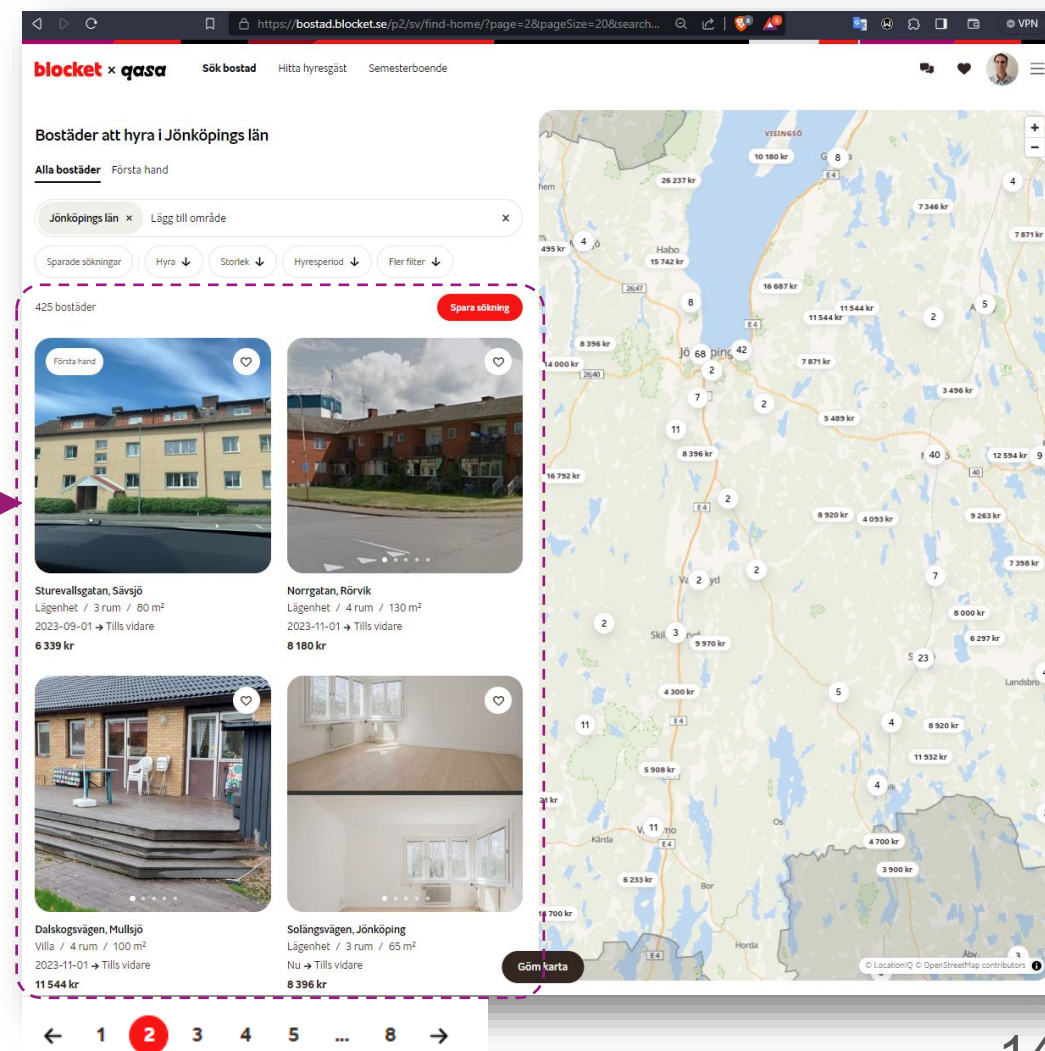
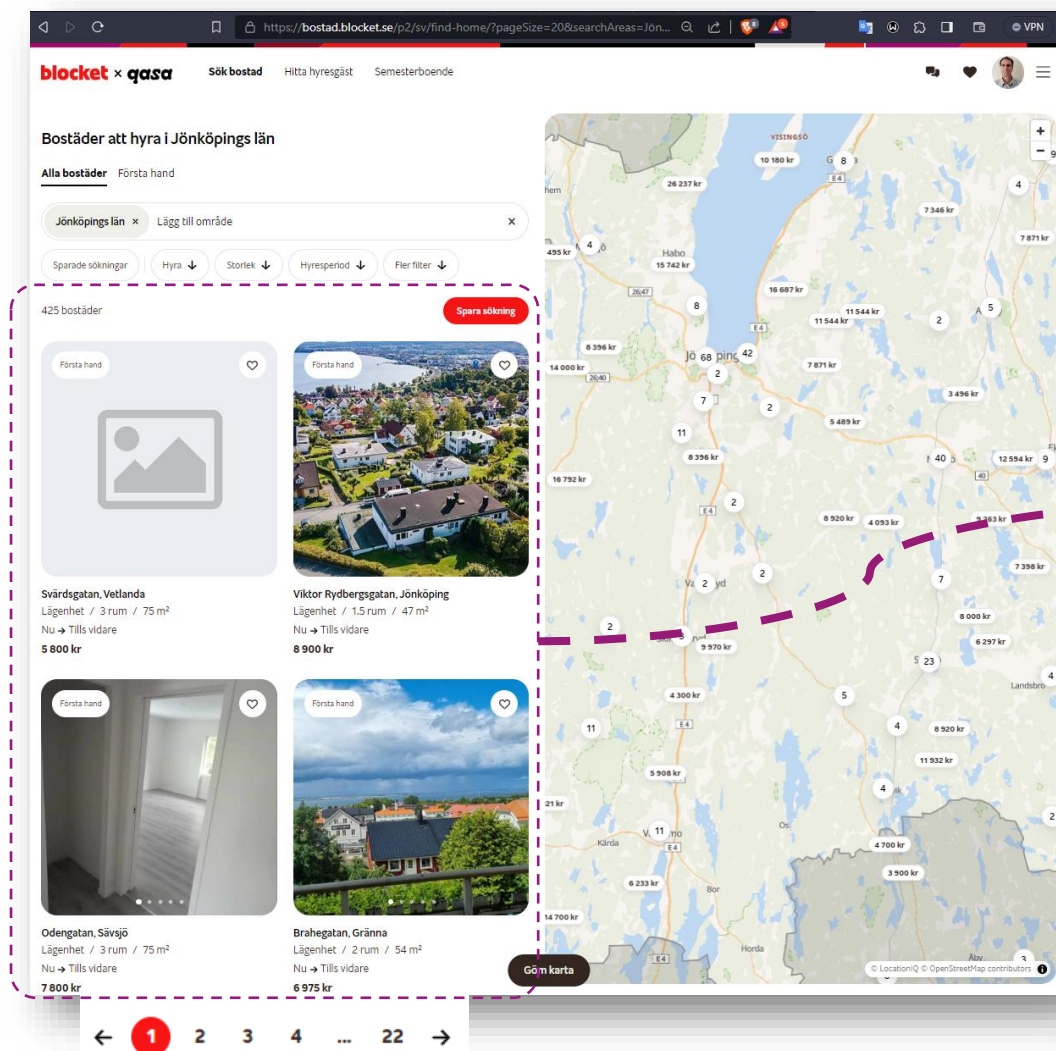
## 3) DEVELOPING DYNAMIC WEB PAGES

- **Dynamic** web pages: more complicated pages (with **data** coming from **databases**)
- We want to load only the information that **changes**

- **Dynamic** web pages: **data** comes from **databases**, load only the information that **changes**!



- **Dynamic** web pages: **data** comes from **databases**, load only the information that **changes!**



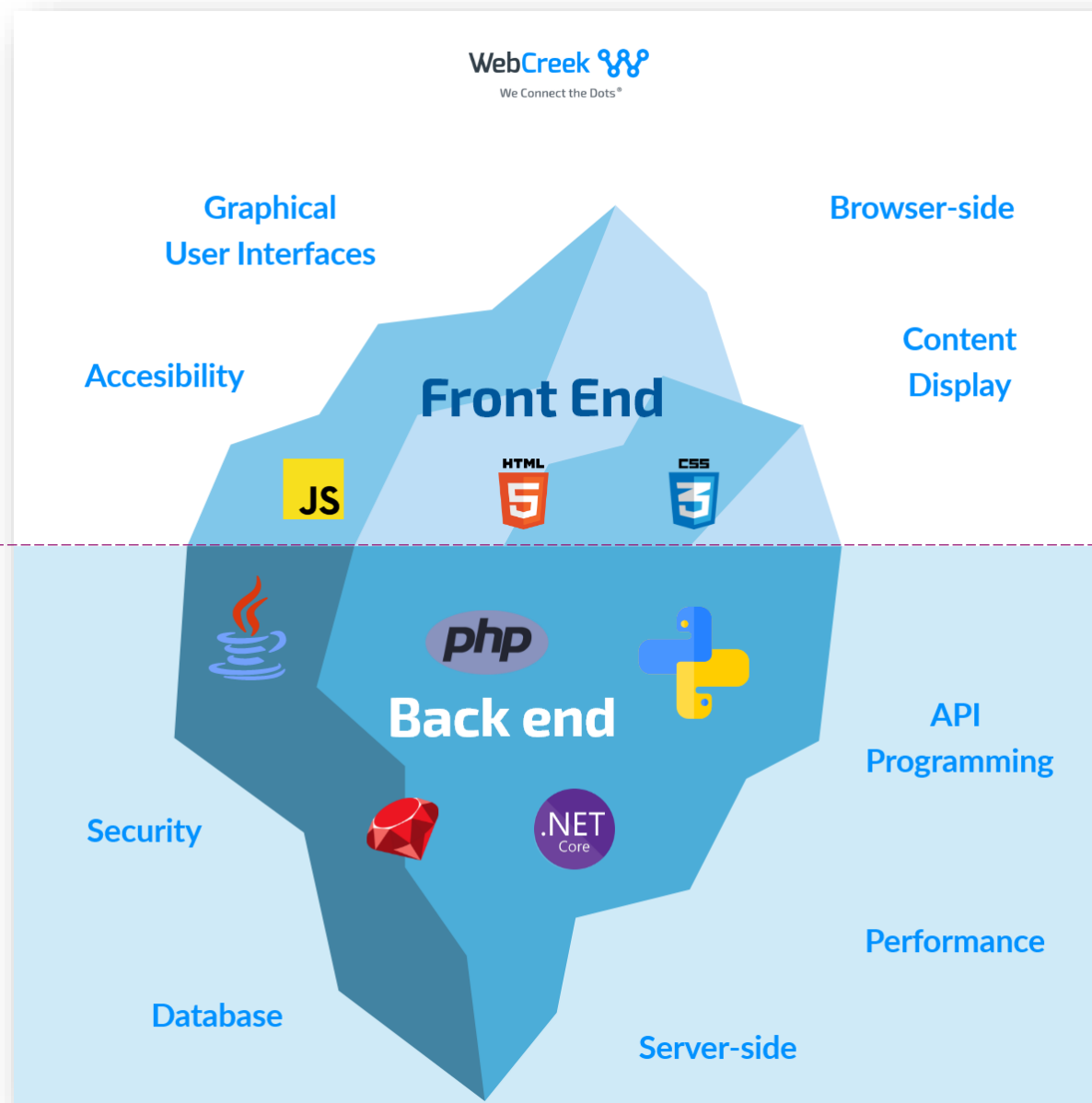


## 4) FRONT-END / BACK-END DEVELOPMENT

- In **web development**, there are three "types" of developers:
  - **Front-end developers**
  - **Back-end developers**
  - **Full-stack developers**
- **Front-end** is the part of the web site the user sees: HTML, CSS, Javascript (client-side)
- **Back-end** is the part of the web site the user doesn't see: Javascript (server-side), PHP, Ruby On Rails, Python (Django), Java Server Pages (JSP), Microsoft DotNet...
- **Full-stack** is the development in both client-side and server-side

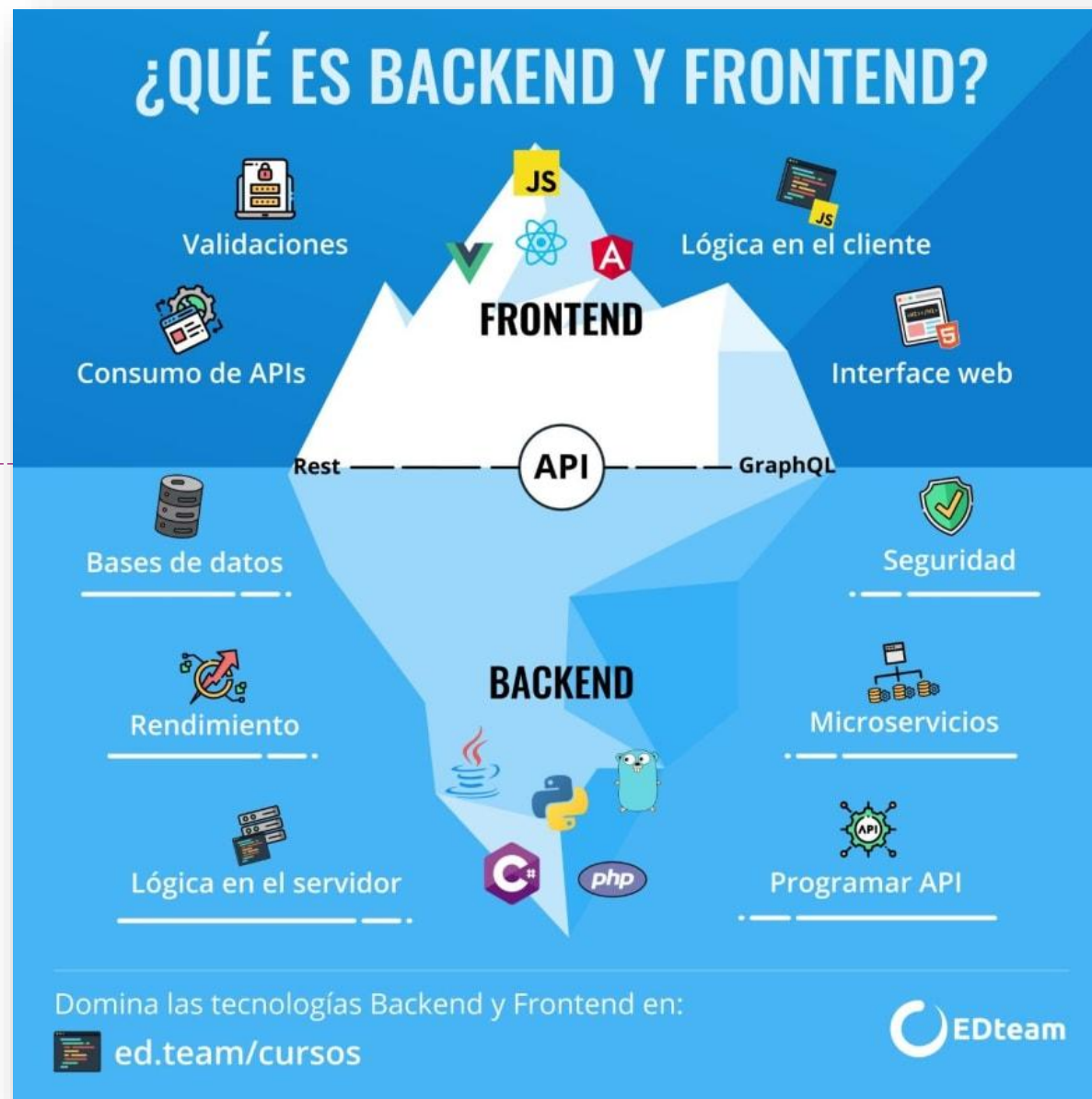
**Client-side (runs  
in the browser)**

**Server-side (runs  
on the server)**



Client-side (runs  
in the browser)

Server-side (runs  
on the server)



**Client-side  
(runs in the  
browser)**

**HOW THINGS LOOK  
TO THE USER**

## FRONT-END

*(Images, Content & Structure)*

Markup and web development languages such as JavaScript, HTML & SCC

Front end frameworks like React, React Native, Flutter, Angular

Asynchronous requests & Ajax

Web editing software

Image editing

Accessibility

Cross-browser issues

SEO

Angular  
JavaScript  
ReactJS

**Server-side  
(runs on the  
server)**

**HOW THINGS  
WORK**

## BACK-END

*(Business, Logic & Data)*

Programming & scripting such as Python, Java, Node.js, Ruby and Perl

CMS

Server architecture

Database administration

Data transformation

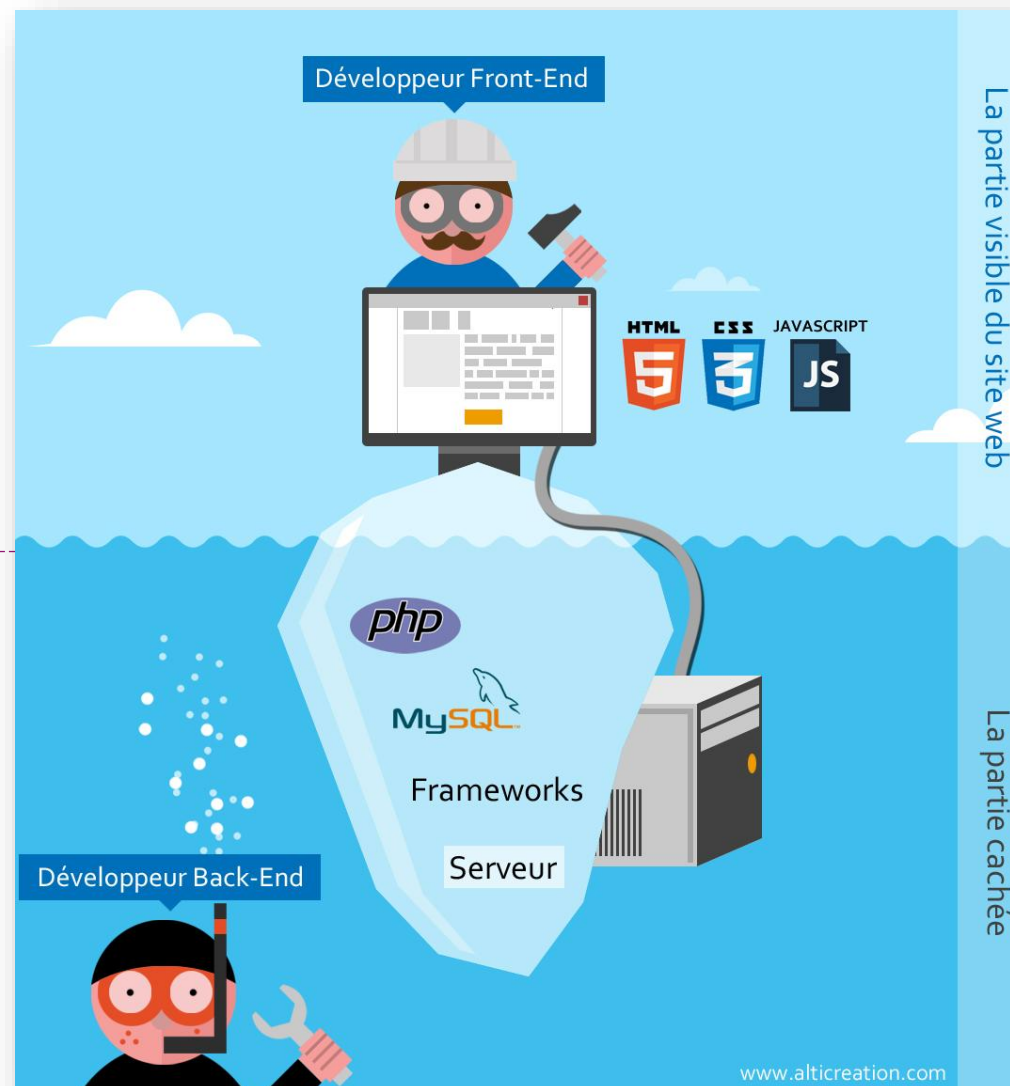
Scalability

Security

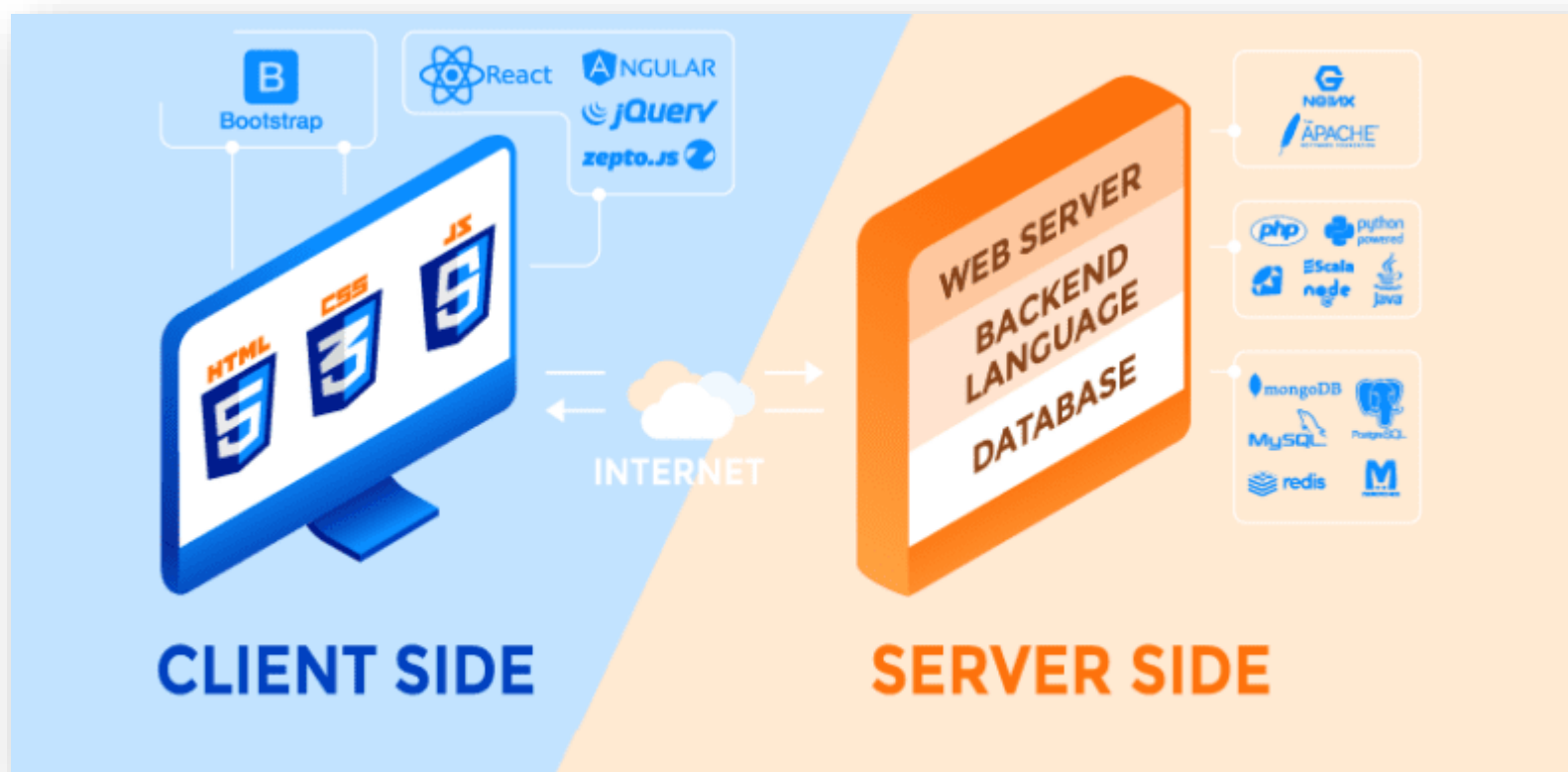
RoR  
PHP  
MySQL  
Node.js  
Python  
MongoDB  
WordPress

**Client-side (runs  
in the browser)**

**Server-side (runs  
on the server)**



## 4) FRONT-END / BACK-END DEVELOPMENT





## 5) WHY NODE.JS? WHY JAVASCRIPT?

- Node.js is a **full-stack development tool**
- It **runs on the server-side**
- However it can use **front-end(client-side) libraries, CSS frameworks**, access to **databases** directly on the server-side

```
const http = require('node:http');

const hostname = '127.0.0.1';
const port = 3000;

const server = http.createServer((req, res) => {
  res.statusCode = 200;
  res.setHeader('Content-Type', 'text/plain');
  res.end('Hello, World!\n');
});

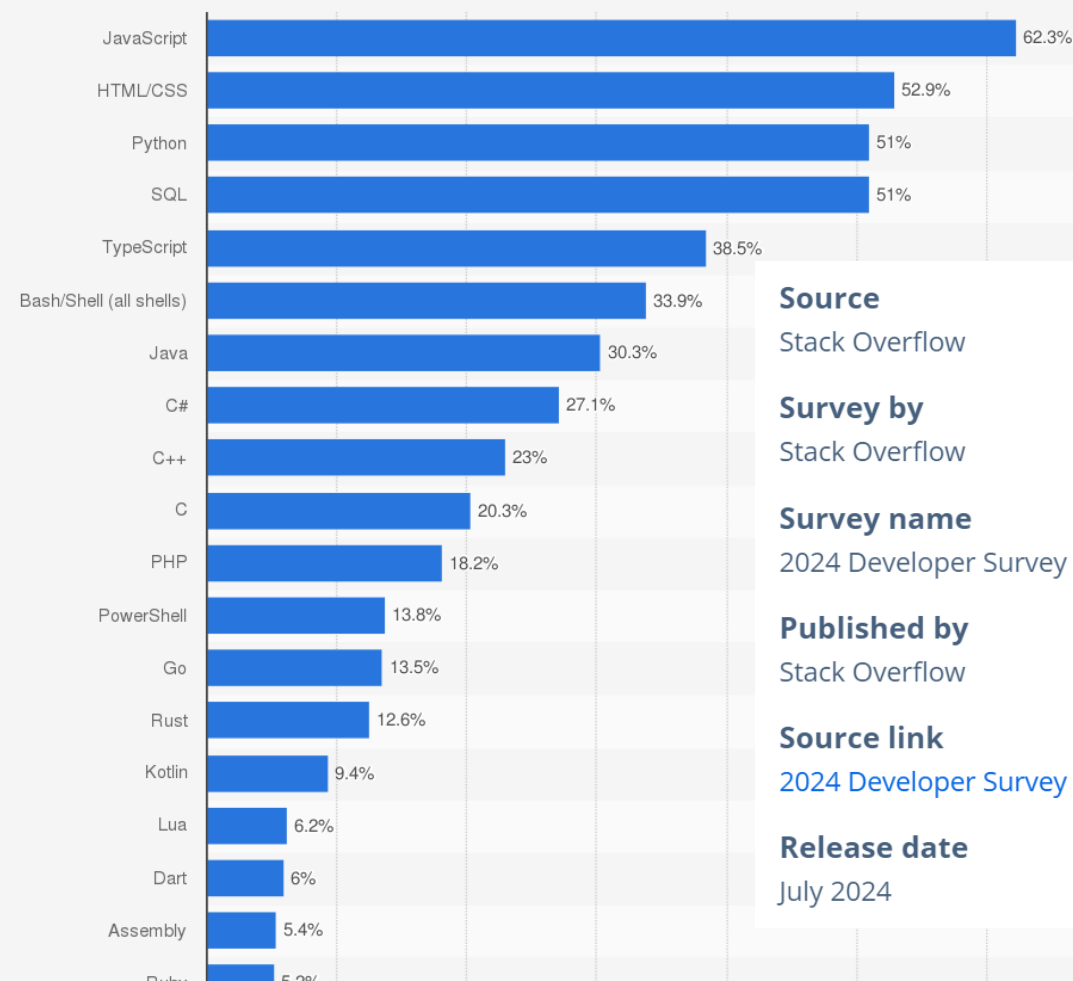
server.listen(port, hostname, () => {
  console.log(`Server running at http://${hostname}:${port}/`);
});
```

## 5) WHY NODE.JS? WHY JAVASCRIPT?

- **Javascript** is a very important programming language both on the client and the server side
- It is an **asynchronous event-driven, object-oriented, dynamic typing** language

Source: <https://www.statista.com/statistics/793628/worldwide-developer-survey-most-used-languages/>

Most used programming languages among developers worldwide as of 2024



**Source**

Stack Overflow

**Survey by**

Stack Overflow

**Survey name**

2024 Developer Survey

**Published by**

Stack Overflow

**Source link**

[2024 Developer Survey](https://www.stackoverflow.com/survey/2024)

**Release date**

July 2024

# 5) WHY NODE.JS? WHY JAVASCRIPT?

- **Node.js (Javascript on the server-side)** is a **full-stack** programming language (both on the client and the server side)
- Many companies are looking for Node.js/Express programmers!

## Om jobbet

Vi söker dig som har erfarenhet av Phyton, Rejact, MongoDB, Java, SQL,

- God erfarenhet av Angular, Typescript, RxJS, Node.js samt gärna C# och api-utveckling med .NET, Azure Services, T-SQL och agila arbetssätt såsom Scrum.
- Meriterande med kunskaper inom UI/UX och design, apputveckling med Kotlin/Swift, SSR med Angular Universal, Entity Framework, ElasticSearch, Redis, Varnish, Docker, Kubernetes, Rancher, SQL Server och datamodellering.
- Erfarenhet av att delta i eller driva projekt och workshops.
- Erfarenhet av att utbilda och informera.
- Minst två års arbetslivserfarenhet från liknande roller.

## Vår tech stack:

- Javascript / Typescript
- NodeJS
- React
- Remix.run
- Docker
- MongoDB
- ElasticSearch
- GraphQL

## Kompetenser

### Meriterande

- ASP.NET, programmeringsspråk
- C#.NET, programmeringsspråk
- Visual Basic.NET, programmeringsspråk
- PHP, programmeringsspråk
- Visual Studio.NET, programmeringsverktyg
- ASP, programmeringsspråk
- HTML, programmeringsspråk
- CSS, programmeringsspråk
- JavaScript, programmeringsspråk

- Developing web applications with ReactJS or VUE
- Building scalable and performant APIs using Node.js
- Working with cloud services such as AWS or Google Cloud Platform

The screenshot shows the Arbetsförmedlingen website with search results for 'fullstack programmerare'. The results list several job openings with details like company name, location, and publication date.

| Job Title                        | Company                           | Location  | Publication Date                 | Action |
|----------------------------------|-----------------------------------|-----------|----------------------------------|--------|
| Fullstack programmerare (2 jobb) | Hantverkshjälpenonline Sverige AB | Stockholm | Publicerad 15 augusti, kl. 11.04 | Spara  |
| Systemutvecklare fullstack       | Site Server Sverige AB            | Östersund | Publicerad 21 augusti, kl. 14.36 | Spara  |
| Fullstack-utvecklare (7 jobb)    | Tillväxtverket                    | Östersund | Publicerad 21 augusti, kl. 13.40 | Spara  |
| Fullstack-utvecklare             | Proccruitment                     | Karlstad  | Publicerad 21 augusti, kl. 10.44 | Spara  |
| Fullstack Developer              | Kundo                             | Stockholm | Publicerad 14 augusti, kl. 16.50 | Spara  |

**REMEMBER:**

**WEB DEVELOPMENT FUNDAMENTALS**

**WEB      DEV   IS   FUN**

# ANY QUESTIONS?